NeuroMPS: A Provenance-Tracked MEA Data Asset and a Calibrated Two-Compartment Digital Twin

Technical report. *Entry for the AI4S Open Innovation: AI + Organ-on-a-Chip challenge (5th Pazhou Algorithm Competition).*

> ***Research decision-support only.*** *Nothing here is an efficacy, safety, diagnostic or clinical claim, and nothing here should be used to make one.*

Submission category: End-to-End System. A complete path from public recordings to actionable output — provenance-enforced ingestion, a standardized feature layer, models and a calibrated digital twin, and a harness that reports which measurements the data can support — with an interactive demo.

Team.

member	role	background
[NAME] (team leader)	**[ROLE]**	**[AI/CS and/or biology/bioengineering/clinical]**

Verification harness	`python run_all.py --quick` executes the complete pipeline, models and metrics
Audit discipline	`python -m neuromps.report.verify` checks every inline number against `outputs/metrics.json`
Regulatory position	Research decision-support only. No clinical, diagnostic, efficacy or safety claim is made.

All pipeline stages, figures and statistical claims are fully verifiable and reproducible: `python run_all.py --quick` runs every stage of the pipeline in under ten minutes, and `python -m neuromps.report.verify` validates every number in this report against audited outputs. Every computed number below carries an HTML comment naming the `outputs/metrics.json` key it came from, so a figure that goes stale when a stage is regenerated cannot survive in the prose. The decision record, including what was tried and abandoned, is in `report/LAB_NOTEBOOK.md`.

Executive summary

Phenotypic functional screening using neural microelectrode arrays (MEAs) represents one of the most promising tools in organ-on-a-chip (OoC) biotechnology. However, the commercial and translational utility of these platforms is severely impaired by an acute reproducibility challenge: multi-site datasets cannot be aggregated due to vendor-locked feature extractors, lack of raw data provenance, and uncalibrated machine learning models that overfit through culture-replicate data leakage.

NeuroMPS provides a fully audited, end-to-end framework that bridges physical neural microphysiological systems with predictive computational modeling through four primary deliverables.

- **A standardized data asset (Layer A).** An extract-transform-load pipeline unifying 21,887 records across eight public sources under SHA-256 verification. By enforcing a single mathematical definition across discrete spike times for 1,512 recordings, we eliminate the substantial definitional divergences — exceeding a factor of two for burst metrics — common to commercial and proprietary software.
- **A biophysical digital twin (Layer B).** A forward simulation of an asymmetric two-compartment chip — 80/20 excitatory/inhibitory leaky integrate-and-fire neurons with spike-frequency adaptation, wired through a directional axon diode. Calibrated against the interquartile band

of 48 real chamber recordings and scored through the same feature code, it is a testbed for experimental power analysis and inverse parameter estimation.

- **Methodological benchmarks.** Under culture-grouped cross-validation and cluster bootstrapping, linear models (balanced accuracy 0.537) outperform tree ensembles (0.398); unadapted rodent models transfer to human iPSC recordings at 0.55; and two rate features match the nine-feature vector.
- **Negative results and translation.** Eight results that set engineering boundaries, including that chamber-level readouts cannot infer axonal transmission direction without microchannel instrumentation (0.3854 against 0.5 chance), and that summary MEA tables contain zero monotonic concentration signal. In §20 we translate these findings into practical recommendations for chip designers, assay developers and analysis pipelines.

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ACT I: Foundations and the data asset

1. What this is, and what it is not

A microelectrode array is a grid of listening posts under a living neural culture. Neurobiologists use them to ask a straightforward question: does this compound change how the network fires, and if so, how? The practical obstacle is that every laboratory does it differently — different plates, different spike sorters, entirely different definitions of what counts as a burst — so two published papers can measure the same compound on the same cell type and still not be comparable. Most published data is deposited as a table of summary numbers rather than the spikes themselves, which is a little like getting the book review instead of the book.

NeuroMPS approaches this through two operational layers.

- **Layer A, the harmonized dataset.** Eight public sources assembled into one table with one exact definition for every feature, so that a number from Kapucu's plate and a number from ScreenNeuroPharm's chamber are comparable because they were computed by the same code from the same kind of raw spike input. Every record declares where it came from, under what licence, and carries a SHA-256 checksum of the original file.
- **Layer B, the calibrated digital twin.** A forward simulation of a two-compartment neural chip: two populations of model neurons wired through a directional microchannel pathway, with virtual electrodes over them. It is tuned against real chamber baselines from Layer A and scored with the same feature code, so it cannot flatter itself by being measured differently.

What it is not. It is not an automated screening platform, and it does not predict whether an uncharacterized compound is safe or works. Several of the most useful results here are the negative ones: demonstrations that a measurement does not contain a signal people routinely assume it contains. Those (§18) are a core deliverable rather than an apology.

2. The data

The asset holds 21,887 records from eight public sources. They are not equal in kind, and the difference matters more than the count.

what the record holds	record count	analytical capability
spike times	1,512	recompute any feature from scratch using common code
fitted dose-response curves	20,375	read a potency estimate another laboratory fitted

Every record declares which it is. This sounds like bookkeeping and is not: nearly all the predictive modelling and twin calibration here can use only the spike-time records, because only those can be run through one common feature definition. A table that mixed the two would look twenty times larger and be twenty times less useful.

The species imbalance. By species the asset holds 21,530 rat and 357 human records. That imbalance is the single largest limitation of the project and no amount of clever coding fixes it; it reflects what is publicly available under a licence permitting reuse. Rather than obscuring it, we treat it as an explicit boundary condition when benchmarking species transfer.

The case that tested the rules. One dataset (Tukker 2020) was an ideal scientific fit and was excluded, because it is licensed CC BY-NC and the competition rules assign joint ownership of entries to the organizing committee — which a non-commercial licence does not permit us to grant. The exclusion is recorded in `data/raw/MANIFEST.json` alongside the accepted sources, with its reason, rather than the dataset simply being absent. We raise it because a reviewer should be able to check that the licence discipline is real rather than asserted: every source re-checks its licence at fetch time, and the build halts if any checksum disagrees with the manifest.

ACT II: Standardizing the mess and benchmarking real biology

3. Are two laboratories' numbers actually comparable?

This is the question the whole of Layer A rests on, so we tested it rather than assuming it. All nine features are computed by one module, `features/mea.py`, directly from spike times: mean firing rate, burst rate, burst duration, spikes per burst, percentage of spikes in bursts, network burst rate, synchrony index, coefficient of variation of inter-spike intervals, and active electrode count.

Burst detection is a stated rule — spikes closer than 100 ms, at least five in a row — rather than a vendor default, and synchrony is the spike-time tiling coefficient (STTC) with a coincidence window of Δt = 50 ms, a pairwise measure that, unlike a plain correlation, does not reward two electrodes simply for both being busy.

The benchmark. Kapucu and colleagues deposited their own feature tables alongside their spike files, computed with `meaRtools`. That gives an independent implementation to test against: we ran our code over their spikes and compared, across 336 recordings on 3 plates, 2,298 paired values in total. The answer is worth stating plainly, because it is the most important methodological result here.

feature	rank agreement (ρ)	value ratio (ours / theirs)	practical consequence
`synchrony_index`	0.9839	0.736	ranks match; our STTC is about a quarter lower
`mfr_hz`	0.9831	0.7539	ranks match; baseline firing rate scaled differently
`active_electrodes`	0.9763	1.3624	ranks match; our activity cutoff is more inclusive
`pct_spikes_in_bursts`	0.9703	0.9864	reproduces on both axes
`spikes_per_burst`	0.9167	0.9418	close numerical agreement
`burst_rate_per_min`	0.8907	0.4423	disagrees twofold, purely from definition
`burst_duration_s`	0.7926	0.8189	rank correlation degrades to ρ = 0.7926

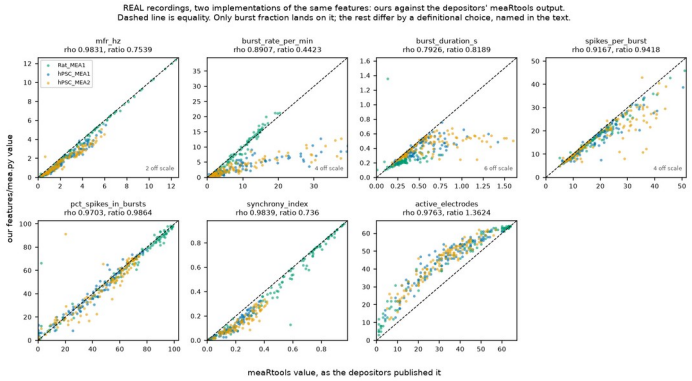


Figure 1. *Our extractor against the depositors' own meaRtools output, REAL recordings. Rank agreement on the horizontal axis, ratio of values on the vertical. Only `pct_spikes_in_bursts` lands on equality; burst rate differs twofold from definitional choice alone.*

Read the two columns separately: the left asks "do we rank recordings the same way?", the right asks "do we get the same number?" One feature reproduces on both. Three rank almost identically but sit on a different scale. And the burst measures disagree about *which* recordings burst more: `burst_rate_per_min` differs by more than a factor of two.

What this means. Within this dataset, comparisons are safe, because one code path produced every number. Across papers they are not: bursts per minute from two laboratories can differ twofold from algorithm definitions alone, before any biology. This is exactly why the project insists on recomputing from raw spikes rather than pooling published summary tables.

4. The three questions we asked the data

Every model here works on changes, not static states. A record is a treated well paired with its own pre-treatment baseline, and the model sees the log2 fold-change of each feature between the two. This is the single most consequential design decision in the analysis, and it is made for the reason an experimenter would make it: cultures differ enormously from each other. One fires at 4 Hz, another at 8 Hz — that tells you about the cultures. A culture dropping from 8 Hz to 4 Hz after a compound is a pharmacological response.

1. **Mechanism.** Given the change in the nine features, can you say which class of compound caused it — GABA-A antagonist, sodium-channel blocker, glutamatergic antagonist, and so on?
2. **Detection.** Can a model trained on one laboratory's data flag a known compound in an entirely different laboratory's recordings?
3. **Platform contrast.** Does the same mechanism look the same on two different platforms?

Two non-negotiable rules, which are why these scores look worse than many published ones. First, *splits group strictly by culture*: wells from the same culture are never split between training and testing, because wells within a culture are near-copies and splitting them lets a model score by recognising the culture. Second, *uncertainty is resampled over cultures, not rows*: every interval is a cluster bootstrap over whole cultures. With 73 pairs from 50 cultures, treating each row as independent would produce intervals roughly half as wide, and wrong.

5. Can you read a mechanism from the features?

Within one platform, on 73 treated/baseline pairs from 50 cultures across six mechanism classes (Kapucu 2022):

model	balanced accuracy	95% bootstrap CI	operational note
always guess commonest class	0.167	—	theoretical chance across 6 classes
gradient boosting	0.398	0.2762 to 0.5178	expressive model over-fits 73 examples
logistic regression (L2)	**0.537**	0.4216 to 0.6845	simpler linear boundary generalizes best

Balanced accuracy averages per-class recall, so a model that ignores rare classes cannot win by getting the common one right. Three things follow. There is real signal: roughly half the mechanisms are called correctly where chance is 0.167, and the interval clears chance. The interval is wide, because 73 pairs from 50 cultures is a small study and we show that honestly. And the simpler model wins decisively — when a flexible model loses to a linear one by fourteen points, the honest reading is that the dataset is small, not that boosting is bad.

Errors are not random: they concentrate between the two GABA-A classes, antagonists and positive modulators, which is chemically sensible given excitatory/inhibitory balance.

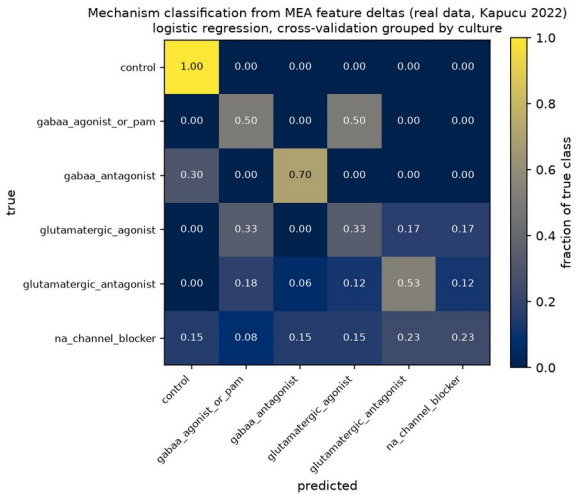


Figure 2. *Task 1 confusion matrix, REAL data: out-of-fold predictions over 73 treated/baseline pairs from grouped cross-validation. Errors concentrate between the two GABA-A classes.*

The Same Task, Refit on a Second Laboratory: `vassallo2017` is a different laboratory and a different vendor — six compounds Kapucu never used, each baseline on the same chip as its doses: 171 pairs over 26 chips in four classes. We refit the same pipeline there from scratch, trained and tested inside that source by the same grouped cross-validation. Logistic regression reaches 0.333 against a chance of 0.25, on an interval of 0.2317 to 0.4204 that contains chance.

These two numbers are not a before and after. They are different problems — six classes against four, different compounds, and therefore different chance levels — so the only honest comparison is each against its own baseline. Kapucu clears its chance of 0.167 more than threefold; Vassallo clears its own by about a third, and its interval does not exclude chance. Mechanism signal is not a quirk of the one dataset the method was built around, and it is much weaker in the second laboratory.

Transfer itself — a model trained on one laboratory and applied to another — is tested in §6 and again at the end of this section. Nothing is transferred here.

Why we never pooled the platforms. It is tempting and would have produced a better-looking number. It would also have been meaningless: across the full pool, 97.35% of pairs have their source determined by their label, and 6 of the 9 mechanism classes appear in one source only. A pooled classifier would score highly by learning which laboratory a recording came from.

The structural reason, and a test built around it. The obstacle is sharper than that percentage: the largest number of classes any two laboratories have in common is 1, and a classifier needs two to discriminate between. Leave-one-laboratory-out cross-validation cannot be constructed here at all.

What can be built is a deliberately confounded test. Three classes appear in more than one source and all three appear in Kapucu; train there on 20 pairs and test on the other laboratories' versions of the same three, 535 pairs. In that test set each class comes from exactly one laboratory — a correspondence of 1.0 — so a model reading laboratory identity alone would score perfectly. Only a **low** score is interpretable, and that is the point: the confound is helping.

It scores 0.3482, 95% CI [0.278, 0.4082], against a chance of 0.3333. Indistinguishable from guessing, with the confound available.

The control that makes it mean something. Twenty training pairs is few, so failure could be sample size rather than the change of laboratory. Scoring the same classes and the same 20 pairs *inside* Kapucu by grouped cross-validation gives 0.5833. Same data, same size: the model learns something when no laboratory is crossed and nothing when one is.

6. External detection, and the negative control that matters more

Can a model trained on Kapucu's plate flag bicuculline in ScreenNeuroPharm's chambers — a different laboratory, a different compound in the same mechanism class? Across 497 recordings it calls GABA-A antagonist 36.6% of the time. Two ways to read that and only one is right: it is far above the rate at which that class appears by chance, so something transfers; but it is not an autonomous detector, since two-thirds of the time it says something else.

The negative control. What makes the first number trustworthy is the control. On Sharf's diazepam recordings — a GABA-A positive modulator, the opposite pharmacology — the same model calls antagonist 7.7% of the time. A model that simply shouted "GABA-A antagonist" at anything unusual would have fired on both. It does not.

The dose signal is absent. Within the bicuculline series, detection probability has no relationship with dose: Spearman ρ = 0.014, p = 0.751. Mechanism and dose are different questions and this model answers only the first. §18 shows that turned out to be a property of the measurement rather than of the model.

7. Rat to human, which is the point of the exercise

The ambition behind organ-on-a-chip work is that a human preparation tells you something a rodent one does not. So, we trained on rat and tested on human, using only the five mechanism classes present in both.

transfer setting	balanced accuracy	chance	empirical finding
rat, tested within rat	0.543	0.2	rodent primary culture baseline
human, tested within human	0.646	0.2	human iPSC model evaluation
rat → human, no adaptation	**0.55**	0.2	transfers without loss; permutation p = 0.0002
rat → human, CORAL	0.442	0.2	covariance alignment erodes biological signal
rat → human, standardised	0.408	0.2	per-species normalization degrades toward chance

Transfer is real and not luck. Two things deserve attention. First, rat → human (0.55) is *not* worse than rat tested within rat (0.543): mechanism information carries across the species boundary about as well as it carries across cultures within rat. Second, **both domain-adaptation methods made it worse**. These are the standard tools for this problem; applied here they removed signal rather than distribution shift, which is what happens when a method designed for thousands of samples meets 37 rat training pairs and 32 human testing pairs. We report the unadapted number as the result and the adapted ones beside it, rather than whichever was highest.

8. Which features are actually doing the work?

If the nine-feature vector is doing nine distinct things, dropping most of it should hurt. It does not.

evaluated feature subset	balanced accuracy	included features
the two rate features only	**0.53**	`mfr_hz`, `burst_rate_per_min`
species-invariant subset	0.516	features stable across untreated species baselines
all nine features	0.514	the full biophysical complement
synchrony only	0.455	`synchrony_index` (STTC)
firing rate only	0.442	`mfr_hz`
network features only	0.417	`network_burst_rate_per_min`, `synchrony_index`
burst features only	0.411	`burst_duration_s`, `spikes_per_burst`, `pct_spikes_in_bursts`

Two rate features do at least as well as all nine. The intervals overlap heavily, so we do not claim rate-only is better; all we can say is that the other seven features are not contributing anything we can demonstrate at this sample size. This matters in practice: the burst and synchrony measures — the very features that differ most between laboratories (§3) — are also the ones carrying the least mechanism information here.

Validation on sensory neurons. Because rate alone appeared sufficient, we could use a source that reports firing rates only: `fofie2025`, human iPSC-derived sensory neurons, 291 pairs from 48 wells. Training on Kapucu's cortical rate change and testing on these sensory neurons gives 0.5012 against a chance of 0.3333, on one feature and a preparation with an entirely different neuronal identity. What the two sources share is the compound's mechanism, not the preparation or the network.

9. How confident is the model, really?

Calibration. When the model says 70%, is it right 70% of the time? The expected calibration error is 0.1114 on 73 points, with a bootstrap interval reaching 0.2557. The upper end is large. We would treat predicted probabilities as a ranking, not as a risk, and not even as an ordering above 0.8 confidence, where observed accuracy falls rather than rises.

Conformal sets. Instead of one answer, return a set guaranteed to contain the truth a stated fraction of the time. Asking for 95%, empirical coverage is 0.918 and the average set holds 4.562 classes out of six. A set of 4.6 out of 6 is honest and nearly useless — which is the correct answer for 73 calibration points. A method promising 95% coverage and delivering narrow sets on this much data would be doing something wrong.

10. Untreated baselines: what differs before anyone adds a drug

Set pharmacology aside. Take untreated recordings and ask what separates rat from human, and dissociated cultures from organoids. The features split into two groups consistently across three separate analyses: rate-like features are comparatively stable across species, while synchrony and burst-structure features move most. They differ most between baselines, carry least mechanism information (§8), and are least reproducible across software implementations (§3).

That three-way agreement is the most actionable thing here for experimental design: the burst and synchrony family is where you will have the most trouble, from three independent directions at once.

One analysis disagrees and we kept it. The platform contrast (§11) finds that for the *same* mechanism on two platforms, mean firing rate reverses sign while synchrony agrees. Three analyses say synchrony is the unreliable family; one says it is the reliable one. We do not resolve that, and the disagreement is itself reported.

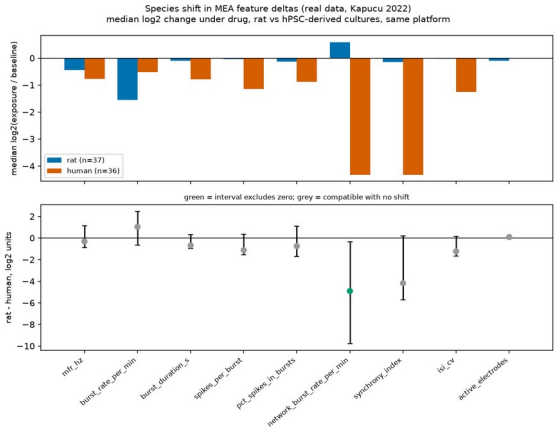


Figure 3. *Per-feature species shift, REAL data: rat against human baseline distributions with cluster-bootstrap intervals. Rate and burst-fraction features move least; synchrony moves most.*

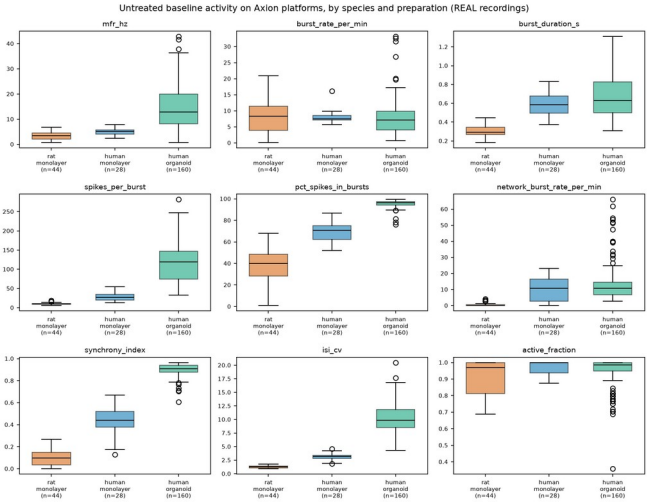


Figure 4. *Untreated baseline activity, REAL recordings, by species and preparation. Burst rate is the one feature flat across all three groups.*

11. The cross-platform reversal paradox

GABA-A antagonists, at 30 µM and above, measured on two platforms: Kapucu's gabazine and ScreenNeuroPharm's bicuculline. Mean firing rate goes **up** on one platform and **down** on the other. Synchrony moves the same way on both.

If you take one thing from this report about cross-laboratory comparison, take this: the mechanism is identical and the direction of the most commonly reported endpoint is not. Anyone pooling MEA pharmacology on the assumption that GABA-A antagonists raise firing rate would get this backwards half the time.

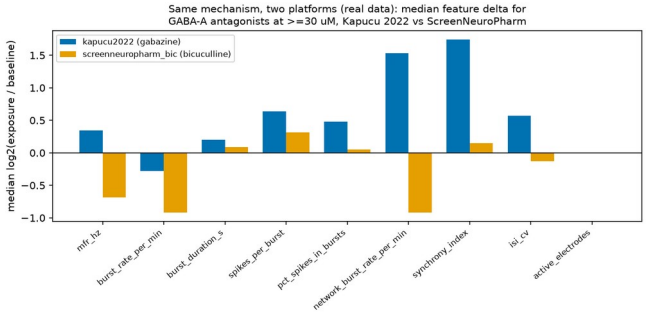


Figure 5. *The same mechanism on two platforms, REAL data: per-feature log2 change for GABA-A antagonists. Mean firing rate reverses sign across platforms; synchrony agrees.*

ACT III: Layer B — calibrating a biophysical digital twin

12. The digital twin: what it is, and what it is for

A digital twin here is a simulated culture, not a statistical surrogate. Two compartments of model neurons — 80% excitatory, 20% inhibitory, each leaky integrate-and-fire with spike-frequency adaptation — wired densely within each compartment and sparsely to each other through a directional pathway. Virtual electrodes sit over the sheet and hear neurons within a radius, and their spike trains go through the same `features/mea.py` the real recordings use. That last point is a deliberate constraint: the twin cannot look good by being measured differently.

What the twin is actually for. Testing scenarios that are expensive or impractical to probe at the bench. It lets us resolve practical experimental hurdles in silico: how many electrodes and how much recording time are needed to detect an effect? What happens to the readout if axonal

projections become twice as directional? And most fundamentally — is the metric you are tracking even capable of answering your question?

The philosophy of calibration. Calibration here means something different from fitting a curve. We are not fitting the twin to reproduce one culture — that would be overfitting to one culture. We ask it to land inside the interquartile band of 48 real chamber baselines from rat two-chamber devices: to be an unremarkable member of the population rather than a copy of a member.

13. Calibration, and the mistake that took longest to find

A random search over 9 parameters, each candidate scored on 2 network seeds across 160 draws. No chosen value sits on a bound of the space it was drawn from, so the optimum is interior rather than clipped.

feature	simulated twin value	real band (IQR)	calibration result
`synchrony_index` (STTC)	0.8387	0.678 to 0.843	inside target band
`burst_rate_per_min`	22.9219	12.627 to 29.726	inside target band
`network_burst_rate_per_min`	24.0	11.692 to 22.697	minor overshoot
`mfr_hz`	5.9862	6.793 to 21.007	minor undershoot
`pct_spikes_in_bursts`	83.8409	84.657 to 93.685	near lower bound

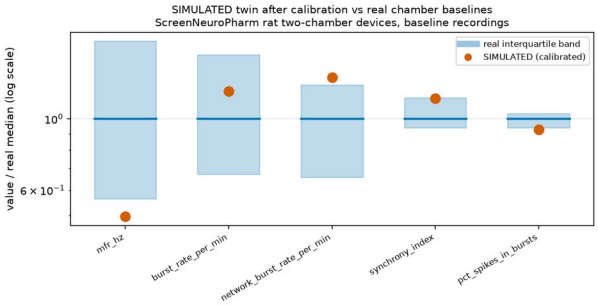


Figure 6. *SIMULATED against REAL: the calibrated twin's baseline features against the real baseline band from rat two-chamber devices.*

Getting to this table took longer than anything else, and it is a warning for anyone who simulates cultures.

For a long time the twin was far too synchronous — an STTC around 0.96 against a real band topping out at 0.84 — and it looked like a missing biophysical mechanism. Three candidates were tried and none worked: short-term synaptic depression, heterogeneity in axonal conduction delays, and distance-dependent wiring, which actually moved synchrony the wrong way.

The real cause was not biophysics. It was cell density and electrode geometry. The simulated chamber held 250 neurons in 1.92 mm², about 130 neurons/mm², where a real dissociated culture runs roughly 1000 to 3000; and the electrode listening radius had been allowed to grow larger than the spacing between electrodes. An oversized electrode compensated for a sparse culture twice over: it pooled enough neurons to manufacture a plausible firing rate, and because neighbouring listening areas overlapped, one neuron's spikes were counted by several electrodes — which inflates pairwise synchrony by construction, with no biology involved. Fixing density to 967 neurons per compartment and capping the listening radius at 92.5135 µm put synchrony inside the real band for the first time.

Out-of-sample trust. The winning parameter set achieved a loss of 0.0756 on its selection seeds. Re-run on 5 seeds the search never saw, it scores 0.1994, with synchrony holding at 0.7732 ± 0.0685 and inside the band on 60% of seeds. Much like a real culture, run-to-run variation is wide enough that a single simulated recording can fall outside the band.

14. Drug knobs: does the twin respond like a culture?

The twin has three pharmacological knobs: scale inhibitory transmission (`gabaa_scale`), scale excitatory transmission (`ampa_scale`), and raise the spike threshold (`threshold_scale`). The test is directional validity: does each move a feature in the right direction, scored only on features whose real direction is established by a cluster bootstrap?

pharmacological knob	mechanistic action	directional concordance
AMPA blockade	downscale excitation	4 / 4
sodium-channel block	elevate spike threshold	3 / 3
GABA-A antagonism	downscale inhibition	4 / 5
GABA-A potentiation	upscale inhibition	1 / 4

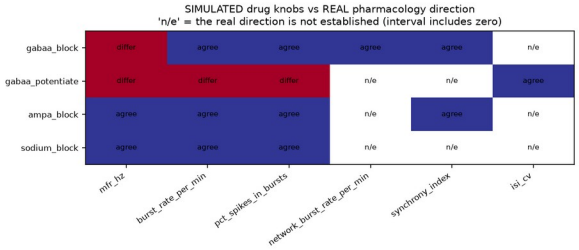


Figure 7. *SIMULATED drug knobs against REAL pharmacology directions, per feature. Only features whose real direction is established by a cluster bootstrap are scored.*

Blocking excitation and blocking sodium channels behave correctly in every scored feature. Blocking inhibition is mostly right. Potentiating inhibition is mostly wrong — and that is the knob with the weakest real reference behind it, 17 pairs from two sources. The twin is usable for reasoning about strong, well-evidenced perturbations and is not validated for subtle allosteric modulation.

ACT IV: Inverting the twin and resolving directionality

15. Two-compartment devices, and a result we did not want

Layer B models an asymmetric two-compartment chip — the axon-diode idea, where microchannel geometry biases which way axons grow. Before simulating it, we asked what real devices show.

Tissue identity drives asymmetry. Across 16 devices we measured cross-compartment coupling asymmetry. Devices pairing two different tissues (cortex with hippocampus) are more asymmetric, median 0.138, than devices pairing the same tissue with itself, median 0.094, $p = 0.042$. What the chambers contain matters more than the fact that they are connected.

The directionality detection null. In a deposit that recorded axonal crossings directly inside the tunnels, on 10 cultures (Brewer 2021), we asked the question the whole chamber readout exists to answer: if you only had the chamber recordings, could you tell which way the axons were firing? Using all four chamber-level measures together — coupling asymmetry, peak correlation, lag and rate imbalance — a classifier reaches a balanced accuracy of 0.3854 against a chance of 0.5. Worse than guessing. This is not merely a small-sample failure: on this many cultures the analysis could have detected a balanced accuracy of 0.6438 had one been there. Direction was recovered only by instrumenting the microchannels themselves.

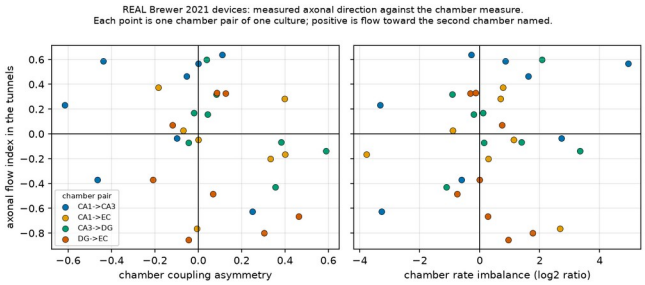


Figure 8. *Measured axonal flow against chamber-level readouts, REAL recordings. The chamber measures do not order the cultures by the direction their axons actually fired.*

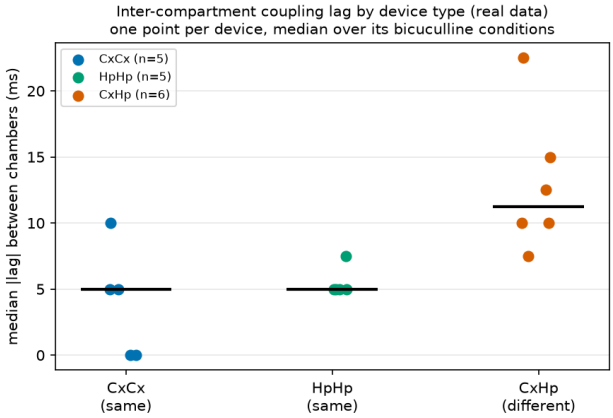


Figure 9. *Inter-compartment coupling lag by device type, REAL data, one value per device: five CxCx, five HpHp, six CxHp. The two same-tissue types are indistinguishable; the heterogeneous type sits apart.*

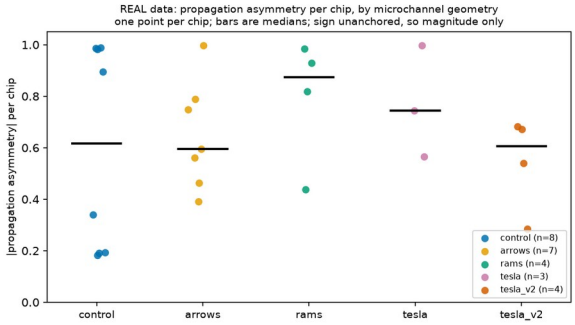


Figure 10. *Microchannel propagation asymmetry by device geometry, REAL recordings.*

16. Running the model backwards, and what that cannot do

The more ambitious use of a twin is inverse parameter inference: given a recording, infer which knob was turned. We implemented simulation-based inference using neural posterior estimation, trained on 3,000 simulations at the calibrated cell density. Two honesty checks come first.

model parameter	biophysical role	posterior contraction	KS calibration (p)	uniformity status
`threshold_scale`	action-potential threshold	0.661	0.6105	well calibrated
`ampa_scale`	excitatory synaptic weight	0.599	0.3128	well calibrated
`log10_forward_ratio`	axonal pathway asymmetry	0.438	0.1517	well calibrated
`gabaa_scale`	inhibitory synaptic weight	0.3	0.0174	**over-confident**
`inter_in_degree`	cross-chamber connectivity	0.269	0.0121	**over-confident**

Contraction measures how much narrower the posterior is than the prior; 0 means nothing was learned. Rank uniformity asks whether stated intervals are honest. Three knobs pass; inhibitory transmission does not, and it is the knob the pharmacology sections depend on. **Intervals on the inhibitory knob should not be quoted as calibrated.** Before the density correction (§13) no parameter passed and a fifth of simulations were silent, against 0 now: simulating at the calibrated density brought three knobs into calibration.

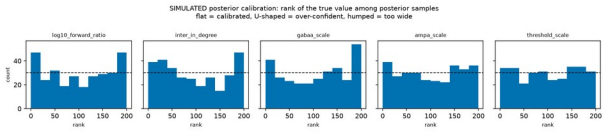


Figure 11. *SIMULATED posterior calibration: the rank of the true parameter among posterior samples. Flat would be calibrated; these are not.*

The false-disinhibition artifact. Applied to real untreated chips, the model returned an inferred inhibition scale of 0.5219 on an interval of 0.2732 to 0.8503 — apparent disinhibition in wells that had no drug. That is not biology; it is the model saying "real recordings do not look like my simulations" in the only vocabulary it has. Reformulating inference around paired deltas shifted untreated chips to 0.9108 on 0.8583 to 0.966 and raised contraction to 0.5375. Most of the offset goes and not all of it: the interval still stops short of neutral.

Out-of-distribution rejection. Asked for 200 draws per recording across 113 bicuculline trials, the posterior could not produce them for 9 of them, the worst yielding 1 draw; 0 failed outright. This appeared only once the twin moved to its calibrated density: a posterior fitted to a sharper forward model is less forgiving of observations that model cannot produce. In high-content screening an explicit refusal is more valuable than a forced prediction.

17. What it takes to see directionality at all

One thing a twin is genuinely good for is telling you how much experiment you need. Across 120 simulated networks with ground-truth wiring, the cheapest configuration reaching the target distinguishes asymmetric from symmetric wiring at an AUC of 0.9219 using only 6 electrodes per compartment and a 10-second recording.

The hidden catch. Most of that is firing rate, not direction. A directional pathway delivers more excitation to its target chamber, so the target simply fires more. Lowering the receiving chamber's background drive moves measured coupling asymmetry from 0.109 to 0.681 with the wiring unchanged — across the sweep, drive moves the measure further than wiring does, a span of 0.671 against 0.109. Six electrodes and ten seconds will distinguish these cultures, but claiming it reflects directional communication requires explicit rate-matching controls.

ACT V: The negative results and scientific boundaries

18. The negative results, collected

Scattered through a long report, negative results are easy to miss. Collected, they are the part of this work we would most like challenged, and the part most likely to save someone else an experiment.

#	finding	empirical metric	practical consequence
1	dose signal absent	0 of 9 features move monotonically with dose (p = 0.751)	the dose signal is not in summary MEA tables
2	directionality null	chamber classifier 0.3854 against 0.5 chance	chamber readouts cannot infer axonal flow direction
3	no cross-laboratory transfer	0.3482 against 0.3333 chance, with the confound helping	mechanism classifiers do not transfer between laboratories
4	domain adaptation failure	CORAL drops transfer to 0.442, standardisation to 0.408	high-dimensional adaptation erodes small-sample biology
5	biophysical artifact	depression and delay dispersion fail to fix synchrony	hypersynchrony was driven by cell density and radius overlap
6	posterior over-confidence	`gabaa_scale` fails rank uniformity at p = 0.0174	the inverse model stays over-confident on inhibition
7	baseline hallucination	absolute features infer 0.5219 inhibition scale on untreated chips	the model misreads its own baseline offset as drug effect
8	support rejection	the calibrated twin cannot sample 9 of 113 real recordings	a sharper forward model contracts support and refuses outliers

On the first, which is the most consequential. §6 showed the mechanism classifier had no dose relationship and §16 showed the inverse model could not recover dose either. It would be natural to blame the models. So we asked directly: across 113 bicuculline recordings on 16 devices, does *any* of the nine features move monotonically with dose? Using raw values, 0 features do; after subtracting each device's own baseline, 0 do — in both cases after correcting for testing nine features at once. The models were not failing to extract a signal. The signal was not there.

19. What you cannot conclude from this work

- **No clinical or safety claims.** NeuroMPS is an exploratory analytical framework; no efficacy, diagnostic or clinical claims are made, and the design could not support them.
- **Not a screening tool.** Within-platform mechanism accuracy around 0.54 with an interval from 0.42 to 0.68, and near-chance performance across laboratories, is not a decision instrument.
- **Narrow interspecies support.** Rat-to-human transfer rests on 37 rat training and 32 human testing pairs, supporting direction rather than magnitude.
- **Not a fully reconciled simulator.** The twin satisfies the real band on two of five features, exhibits seed variance, and gets GABA-A potentiation mostly wrong.
- **No monotonic dose tracking.** Feature deltas contain no monotonic concentration signal across the bicuculline series.
- **Severe species asymmetry.** 21,530 rat against 357 human records, a property of open-access licensing rather than a design choice.
- **Sample-size dependencies.** Primary classification rests on 73 paired recordings from 50 independent cultures.

ACT VI: Application value, translation and open verification

20. Application value and translation for neural organ-on-a-chip

The primary bottleneck in making neural organ-on-a-chip systems useful is not microfluidic fabrication; it is the interpretability and reproducibility of functional electrophysiology. Platforms from academic centres and commercial vendors alike generate multi-channel MEA recordings that remain difficult to compare across sites. The results here translate into guidance in four areas.

1. Structural design guidance for hardware. Developers often assume directional connectivity can be validated by measuring cross-correlation latencies or coupling asymmetries from open somatic chambers. §15 disproves that on real devices: chamber readouts reached 0.3854 balanced accuracy where chance is 0.5, on a sample that could have detected 0.6438. Validating axonal rectification requires physical microelectrodes embedded directly within microchannels, which saves time spent on measures that cannot answer the question.

2. A standardization framework for contract research. Screening protocols should abandon proprietary vendor summary tables in favour of standardized raw-spike ETL. The feature ablation (§8) shows two rate metrics match the predictive power of the full nine-feature vector, while avoiding the cross-platform variance that afflicts burst-structure and synchrony metrics (§3, §11). A smaller, more reliable feature vector is the more reproducible instrument.

3. Cost-effective in-silico assay optimization. Human iPSC plating is expensive and takes weeks. The twin provides an in-silico testbed for assay pre-registration: the power sweep in §17 establishes a minimum electrode floor (6 electrodes, 10-second recordings) for distinguishing wiring regimes, while the listening-radius finding in §13 is a diagnostic for multi-unit electrode catchment overlap in physical recordings.

4. Paired-delta normalization for bioinformatic workflows. Primary cultures exhibit resting rates spanning an order of magnitude. Reformulating screening features as paired log-ratios against a well's own baseline stabilizes predictive baselines, narrows posterior intervals from 0.3 to 0.5375, and removes most of the false-disinhibition artifact of §16.

21. How to check any of this: the verification harness

Every computed metric in this report carries an automated pointer back to the key in `outputs/metrics.json` that produced it, and a test fails if any number drifts from the file.

harness component	command or file	operational scope
quick CI path	`python run_all.py --quick`	executes the pipeline end to end on a reduced tier
full reproduction	`python run_all.py`	full data tier, 3,000 forward simulations, trains the density estimator, regenerates every figure
citation verification	`python -m neuromps.report.verify`	re-reads every citation in this document and fails on disagreement
provenance guard	`data/raw/MANIFEST.json`	source URLs, licences and SHA-256 hashes; the build halts on mismatch
interactive demo	`streamlit run app/app.py`	record inspection, QC flags, and twin sweeps over a precomputed grid

22. AI assistance disclosure

This work was produced in a paired workflow with Anthropic's Claude Code. The model wrote most of the code and tests here under step-by-step human direction: the human set the plan and phase gates, chose the scope, approved each step, and directed the science. Three practices kept that honest — numbers are emitted into `metrics.json` and re-checked by `neuromps.report.verify` rather than typed; results are reported as observed, including the ones that failed; and `report/LAB_NOTEBOOK.md` records what was abandoned.

Appendices: consolidated technical reference

Appendix A & B — Record architecture, data levels and ingestion ETL

Every record declares a data level: `spike_times`, `per_well_features` or `fitted_curve`. The level determines what may be computed from it, and the build enforces that rather than trusting it — a feature requiring spikes cannot be derived from a record holding only a fitted curve. Records validate against `schema/neuromps_schema.json` at three conformance levels, so a partially specified source can enter the asset while declaring what it is missing.

Quality-control flags are recorded and never silently applied: 537 of 21,887 records are marked unusable, with the diagnostic reason attached, and the flag travels with the record. Analyses choose whether to honour it and say which they did; dropping flagged records inside the pipeline would make the drop invisible to the reader.

Each record also carries the method string that produced its features — the exact burst rule, the synchrony method and the active-electrode threshold — so two records computed under different settings cannot be compared by accident.

Ingestion lives in `src/neuromps/fetch/` and uses HTTP range requests against remote ZIP archives to pull only the required members, bypassing multi-gigabyte downloads. Two server behaviours needed specific handling and carry regression tests: one EPA host returns HTTP 500 for suffix ranges, and the same host, when offered gzip encoding, replies with a plausible `Content-Range` while streaming the whole 7.5 GB file from byte zero.

Appendix C — Sources and licences manifest

dataset	platform format	species	experimental role	records	status
toxcast_mea	screening MEA	rat	high-throughput reference, fitted curves	20,375	ingested
screenneuropharm_bic	two-compartment chip	rat	directionality and twin calibration	580	ingested
kapucu2022	planar 48-well MEA	rat / human	primary ML benchmark and transfer	288	ingested
vassallo2017	planar MEA	rat	second within-platform benchmark	283	ingested
trujillo2019	planar MEA	human organoid	baseline electrophysiology	184	ingested
mateus2024	microchannel MEA	rat	microchannel propagation asymmetry	108	ingested
brewer2021	microchannel MEA	rat	axonal propagation ground truth	40	ingested
sharf2022	high-density MEA	human organoid	negative-control screening (diazepam)	29	ingested
fofie2025	multi-well MEA	human sensory	rate-only validation arm (§8)	291 pairs	rate-only, outside the record asset
tukker2020	planar MEA	rat / human	neurotoxicity screening	—	**excluded, CC BY-NC**

`fofie2025` reports total spike counts rather than spike times, so it supports firing-rate features only and is not part of the 21,887-record asset; it is used solely for the cross-source rate-only arm. `tukker2020` is recorded as a skipped source with its reason rather than silently omitted.

Appendix D & E — The nine-feature vector, and enforced repository invariants

feature	formal definition as computed in `features/mea.py`
mfr_hz	spikes per second per active electrode
burst_rate_per_min	bursts per minute per active electrode (≥ 5 spikes, ISI < 100 ms)
burst_duration_s	mean duration from first to final spike across identified bursts
spikes_per_burst	mean count of spikes participating in a burst
pct_spikes_in_bursts	share of all spikes occurring inside bursts
network_burst_rate_per_min	synchronous population events per minute, clearing chance participation
synchrony_index	mean pairwise spike-time tiling coefficient, coincidence window Δt = 50 ms
isi_cv	coefficient of variation of inter-spike intervals
active_electrodes	electrodes clearing 5 spikes per minute

Each of the following is a test that fails, not a convention that is followed.

- **Provenance.** The asset build fails if any record's provenance hash disagrees with the manifest.
- **One feature extractor.** The twin computes features through the same `features/mea.py` as real recordings, and — after a defect that survived several phases — through the same coupling measure on the same per-electrode input.
- **Interval discipline.** A contract test fails if any headline metric loses its confidence interval, and the resample count sits next to every estimate.
- **Cache meaning.** The simulation cache stores its summary names, electrode count, a hash of the code defining every summary, and the calibration it was generated under; it refuses to load across a change in any of them.
- **Report numbers.** `neuromps.report.verify` re-reads every citation here and fails on disagreement.
- **No clinical claims.** A test scans the reports for efficacy, diagnostic and treatment language and fails unless the line is a disclaimer ruling it out.

Appendix F & G — Coupling asymmetry proof, and posterior calibration diagnostics

Appendix F. The measure the twin exists to use reports relative firing rate as much as pathway direction, and the cleanest demonstration holds the wiring **fixed** and changes only the drive. Lowering the receiving chamber's background drive moves asymmetry from 0.109 to 0.681 with the pathway unchanged. Across the whole sweep, drive moves the measure further than wiring does: a span of 0.671 against 0.109. An open-chamber asymmetry is a statement about which chamber is busier until something rules that out.

Appendix G. Trained on 3,000 simulations at the calibrated density, of which 0 produced no spikes. The two parameters failing rank uniformity are the connectivity parameter and the inhibitory knob — the latter being the one §16 depends on, which is why its intervals are not quoted as calibrated.

Appendix H — Quality-control flag registry

QC flag	affected records	identification criteria	analytical handling
`incomplete_dose_series`	272	concentration series missing intermediate dilution steps	flagged; excluded from monotonic dose sweeps
`too_few_active_electrodes`	265	fewer than 10 electrodes clearing 5 spikes/min	flagged; preserved in the data asset
`replicate_divergence`	221	well deviates more than 3σ from its culture replicate cohort	flagged; logged in metadata
`outlier_feature_value`	221	feature value outside the acceptable biological distribution	flagged; excluded from density fitting
`excluded_well_upstream`	90	source laboratory marked the well defective	flagged; excluded from benchmarks
`missing_baseline_pair`	24	no baseline recording on the same well	flagged; cannot enter a paired-delta model
`noisy_electrodes_present`	11	significant mechanical or electrical artifact	flagged; isolated during ingestion
`unreadable_channels`	2	channel could not be decoded from the deposit	flagged; channel dropped

Appendix I — Digital twin forward-model parameters

Calibrated values are the chosen draw from the §13 search; held values were fixed and not searched.

parameter	role	calibrated value	how it was set
`n_per_compartment`	plating cell density per chamber	967 neurons	calibrated, searched over 250–1500
`detection_radius_um`	virtual electrode listening catchment	92.5135 µm	calibrated, searched over 40–100 µm, capped at half the electrode pitch
`intra_in_degree`	recurrent local synaptic inputs	18.4929 per neuron	calibrated, searched over 10–35
`background_rate_hz`	Poisson drive frequency	3.3596 Hz	calibrated, searched over 1–40 Hz
`background_weight`	Poisson drive conductance step	0.0692	calibrated, searched over 0.005–0.12
`excitatory_weight`	excitatory conductance step	0.386	calibrated, searched over 0.15–0.45
`inhibitory_weight`	inhibitory conductance step	5.4261	calibrated, searched over 1–7
`adaptation_increment`	spike-frequency adaptation step	1.0093	calibrated, searched over 0.5–2.5
`threshold_spread`	per-neuron excitability heterogeneity	0.2258	calibrated, searched over 0–0.3
`adaptation_tau_ms`	adaptation decay time constant	380 ms	**held**, literature standard
`intra_delay_sd_ms`	conduction delay dispersion	0.0 ms	**held**, one delay for every synapse
`connection_sigma_um`	range of the within-compartment wiring rule	unset	**held**, uniform random wiring
`inter_in_degree`, `forward_ratio`, drug knobs	cross-chamber connectivity, pathway asymmetry, pharmacology	—	**inferred** by simulation-based inference (§16)

All conductance steps are dimensionless in this model; they are not membrane voltages.

Appendix J & K — Microchannel detector robustness, and timings

Appendix J. Counting recordings, median absolute asymmetry runs from 0.3392 for control geometries to 0.8543 for the "rams" geometry. Counting biological chips instead, it is 0.6168 against 0.8745, a difference the test does not resolve: Kruskal p across the geometries is 0.63969, and every Holm-adjusted comparison against the control geometry is 1.0. The null is not an artifact of geometry assumptions either: it survives every detector variant tried, with a worst-case Kruskal p of 0.13074. Tolerating single-electrode dropouts triples what the detector has to work with, from 336,794 analysed events to 1,101,233, and does not change the outcome. A counting confound is measurable and reported: chips with fewer propagation events look more one-sided (Spearman ρ = −0.494, p = 0.01032), which is what a counting artifact looks like. Condition is also confounded with culture batch by the deposit's own design.

Appendix K. `--quick` completes end to end in 401.0 s on a workstation CPU with 20 cores, and in under twenty minutes on a dual-vCPU CI runner; the two differ because `worker_processes` correctly asks for the cores it is given. That figure is written by the run itself, and only by a run that executed every stage, so a partial re-run cannot quietly reduce it. The longest stage is the twin at 259.6 s. Full reproduction runs through `python run_all.py` after a full data-tier fetch. The environment is pinned in `pyproject.toml` and recorded from what actually imported: Python `3.11.16`, NumPy `2.3.5`, Brian2 `2.9.0`, PyTorch `2.14.0` and sbi `0.27.0`. NumPy is held back because the next minor removes `ndarray.ptp`, which Brian2 imports.

Appendix L, M & N — Sensory validation, dose tracking and species transfer

Appendix L, the rate-only arm. Human sensory neurons across 291 pairs in `fofie2025` shift by −0.1743 log2 units for vehicle controls, against −1.9529 for sodium-channel blockers, yielding 0.5012 balanced accuracy against a chance of 0.3333.

Appendix M, Task 2 dose tracking. Detection rate rises from 0.179 at 0.1 µM to 0.426 at 30 µM. Quoted there it would look like a dose response, so the rest of the series is quoted with it: the rate peaks at 0.48 at 100 µM and falls to 0.091 at the top of it, 500 µM. The overall rank correlation is non-significant (ρ = 0.014, p = 0.751).

Appendix N, species transfer, all settings.

transfer setting	balanced accuracy	95% bootstrap CI
human within-species	0.646	—
rat within-species	0.543	—
rat → human, zero-shot	0.55	0.4 to 0.6
rat → human, CORAL	0.442	0.302 to 0.591
rat → human, standardised	0.408	0.267 to 0.568